ML-DL Ops Lab 2 Worksheet

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***Abstract*—This work presents a modified ResNet-18 tailored for efficient deployment on CIFAR-10. A custom binary dataloader and a redesigned input stem optimized for 32×3232 \times 3232×32 images reduce computational cost to 557.88 MFLOPs. The model achieves 91.88% Top-1 test accuracy within 25 epochs. Training behavior is validated through Weights & Biases diagnostics, including gradient flow and weight update visualizations, confirming effective residual learning and learning-rate scheduling.**

I. Introduction

Deep Residual Networks (ResNets) have revolutionized image recognition by enabling the training of very deep networks. However, standard implementations are often optimized for ImageNet (224×224), making them suboptimal for smaller datasets like CIFAR-10. Direct application results in excessive downsampling and wasted computational resources.

This report details a complete training pipeline designed to address these issues. The contributions of this work include:

- A custom `Dataset` implementation that parses raw Python binary files directly.
- An architectural modification to ResNet-18 reducing FLOPs by $\approx 70\%$.
- A diagnostic analysis of training dynamics, specifically gradient propagation and weight update flows.

II. Methodology

A. Custom Data Pipeline

A custom `torch.utils.data.Dataset` class was implemented to handle the CIFAR-10 binary batches. Rather than using pre-processed wrappers, the raw bytes were loaded via `pickle`, reshaped to $(N, 3, 32, 32)$, and transposed to channel-last format for compatibility with PIL-based augmentations.

- **Train Set:** 50,000 images.
- **Test Set:** 10,000 images.
- **Augmentations:** Random Crop (32, padding=4), Random Horizontal Flip.

B. Architectural Modifications

The standard ResNet-18 input stem consists of a 7×7 convolution (stride 2) and a 3×3 MaxPool (stride 2). For 32×32 images, this reduces spatial dimensions to 8×8 before the first residual block, destroying fine-grained features.

The architecture was modified as follows:

1) The first layer was replaced with a 3×3 convolution (stride 1).
2) The MaxPool layer was removed entirely.

This maintains the 32×32 resolution through the initial layers, preserving information flow.

C. Training Setup

- **Optimizer:** SGD (Momentum 0.9, Weight Decay $5e^{-4}$).
- **Scheduler:** Cosine Annealing (decaying LR from 0.1 to 0).
- **Batch Size:** 128.
- **Epochs:** 25.

III. Experimental Results

A. Quantitative Performance

The training process was tracked using Wandb. The final metrics obtained after 25 epochs are presented in Table I.

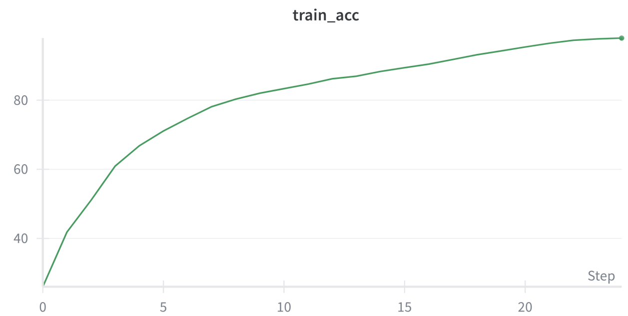
TABLE I: Final Run Metrics (Epoch 25)

Metric	Value
Test Accuracy	**91.88%**
Training Accuracy	97.97%
Final Training Loss	0.0633
Parameters	11.17 M
Inference Cost (FLOPs)	**557.88 M**

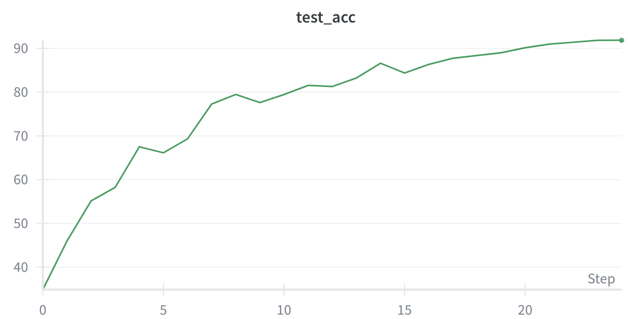
B. Performance Curves

The training dynamics are visualized in Fig. 2. The model shows rapid convergence in the first 10 epochs, with fine-tuning occurring as the learning rate decays.

Fig. 1: Training Loss Convergence. The loss decreases smoothly to 0.0633.

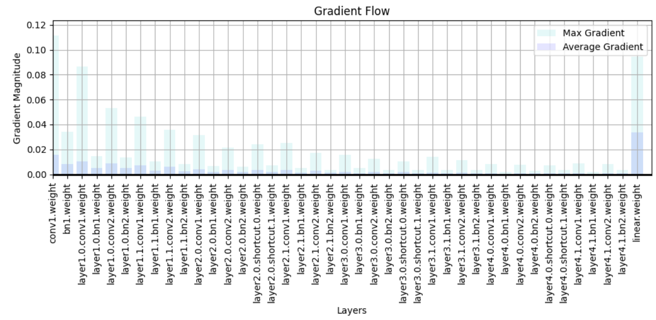


(a) Train Accuracy

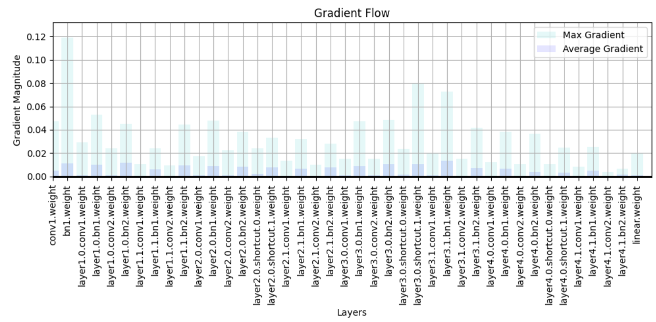


(b) Test Accuracy

Fig. 2: Accuracy Metrics. The model reaches 91.88% test accuracy (right) while maintaining high training accuracy (left).



(a) Epoch 0: Initial Gradients



(b) Epoch 24: Final Gradients

Fig. 3: Gradient Flow Comparison. Healthy gradients are observed throughout the training process.

IV. Discussion and Analysis

A. Gradient Flow Visualization

To verify the network's health, gradient magnitudes were visualized across layers. Fig. 3 compares the gradients at the start and end of training.

- **Finding:** Gradients were non-zero and consistent across the entire depth of the network at Epoch 0 (Fig. 3a).
- **Implication:** This confirms that the residual skip connections effectively mitigated the vanishing gradient problem.

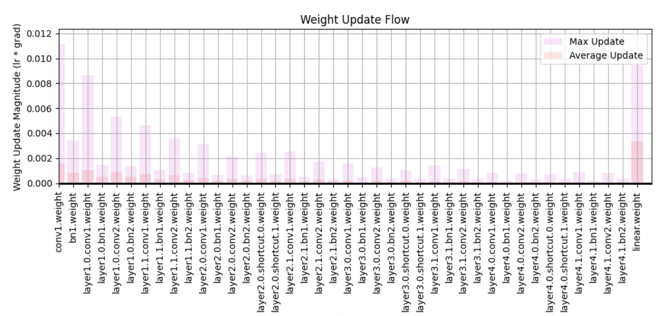
B. Weight Update Flow

The magnitude of weight updates ($\eta \cdot \nabla w$) was monitored (Fig. 4).

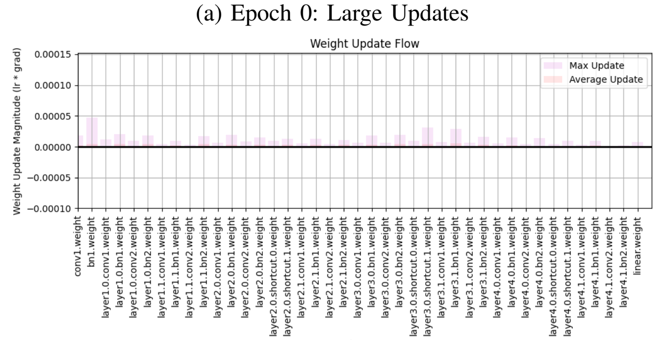
- **Finding:** The updates show a monotonic decrease.
- **Implication:** Large updates at Epoch 0 drive feature learning, while negligible updates at Epoch 24 (Fig. 4b) confirm that the Cosine Annealing scheduler successfully reduced the learning rate to fine-tune the weights.

V. Conclusion

ResNet-18 was adapted for CIFAR-10, achieving **91.88% accuracy** with an efficient **557 MFLOPs** footprint. Custom data handling and architectural modifications were robust, while diagnostics confirmed healthy gradient flow and stable convergence.



(a) Epoch 0: Large Updates



(b) Epoch 24: Minimal Updates

Fig. 4: Weight Update Magnitude. The decreasing trend aligns with the LR scheduler.